PADHAKU
Exploration · Class 9 Science



CHAPTER 6

How Forces Affect Motion

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples; write your own.

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🤔 Think It Over

1 **Why does a canoe move forward when the canoeist pushes water backwards with their paddle, and why does it move faster when they push harder?**

✔ ANSWER

The paddle pushes the water backwards; by **Newton's third law** the water pushes the paddle (and canoe) forward with an equal force.

Pushing harder gives a larger forward force, so by the **second law** the acceleration is larger and the canoe moves faster.

2 **The same canoeist uses the same paddle force in two canoes, one empty and one carrying another passenger. In which case will the canoe move faster?**

✔ ANSWER

The **empty canoe** moves faster.

For the same force, a smaller mass gets a larger acceleration (a = F/m).

⭐ Pause and Ponder · 6.2 Balanced and Unbalanced Forces

1 **A weightlifter lifts a barbell (Fig. 6.8). List two forces that are acting on the barbell. Are**

these forces balanced if the weightlifter keeps the barbell steady?

✔ ANSWER

- Gravitational force (weight) of the barbell, acting downwards.
- Upward force applied by the weightlifter's hands.

When the barbell is held steady (at rest), the two forces are **balanced**: equal in size and opposite in direction, so the net force is zero.

2 **Two players R and S are in an arm-wrestling match (Fig. 6.9). At the instant when the arms tilt to the front direction (out of the page towards you), are the forces exerted by the players balanced? If not, which player exerted the larger force?**

💡 SAMPLE ANSWER

R pushes towards S and S pushes towards R. When the arms tilt towards you (not towards R's side or S's side), neither hand moves along the R–S line.

So the forces of R and S along that line are **balanced** (equal and opposite); **neither** player exerted a larger force at that instant. (check the picture in your book)

⭐ 6.3 The Force of Friction

What if... **What if the force of friction disappears in the world? How will the motion of objects be impacted?**

💡 **SAMPLE ANSWER**

A moving object would never slow down by itself; it would keep moving with the same velocity.

But we could not walk, hold things, or stop vehicles, because friction helps us grip and brake. Objects would slip and slide everywhere.

Activity 6.1 **Let us investigate: a stack of 4 coins is launched by a rubber band stretched to the same mark C on a wooden table, a laminated table, and a polished marble/tile floor. What do you observe and conclude?**

✔ **ANSWER**

Observation: After leaving the rubber band, the stack slows down gradually and stops. It travels the shortest distance on the wooden table, more on the laminated table and the most on the polished marble/tile floor; its velocity decreases more slowly on smoother surfaces.

Conclusion: The force of friction acts opposite to the motion and stops the coins. Friction is different for different surfaces: smoother surface → less friction → larger distance.

Activity 6.2

Let us measure: a wooden block is pulled with a spring balance on each surface. What does the reading when the block just starts moving indicate? If the velocity of the block is neither increasing nor decreasing, what is the net force? Compare the readings.

✔ ANSWER

The reading when the block just starts moving shows the force needed to overcome friction.

If the velocity is constant, the **net force is zero**: the spring's pull equals the force of friction, so the reading gives (approximately) the force of friction.

The readings are different for different surfaces: **smallest** on the surface where the coins went farthest (polished floor) and **largest** where they went the least distance (wooden table).

Think as a Scientist

Thought experiment: if the force of friction between an object and a horizontal floor is zero and you repeat steps 3 and 4 of Activity 6.1, will the velocity of the object

decrease? Will it ever come to rest?

✔ ANSWER

No force acts against the motion after release, so the velocity will **not decrease**.

The object will **never come to rest**; it will keep moving with constant velocity forever (as Galileo argued).

⭐ Pause and Ponder · 6.4 Newton's First Law of Motion

3 **An object is moving with a constant velocity. Is there a net force acting upon it?**

✔ ANSWER

No. By Newton's first law, constant velocity means zero acceleration, so the net force on it is zero.

4 **Suppose no net force is acting on an object. Which of the following situations are possible?**

(i) Object remains at rest if at rest.

(ii) Object keeps moving with a constant velocity if already moving.

(iii) Object is moving with a constant acceleration.

✔ ANSWER

(i) and (ii) are possible.

(iii) is not possible: acceleration needs a net force.

5 **In the real world, it is difficult to find a situation where no forces act on an object. But by applying additional forces, a condition can be achieved where the net force is zero. Explain with an example.**

💡 **SAMPLE ANSWER**

Example: a box is pushed along the floor. Friction acts backwards on it.

If we push forward with a force exactly equal to the friction, the two forces balance and the net force is zero.

Then the box moves with a constant velocity, as Newton's first law says. (Another example: a book on a table — its weight is balanced by the upward force of the table.)

⭐ 6.5 Newton's Second Law of Motion

Activity 6.3 **Let us experiment: a cart is pulled across a table by a thread attached to a hanging cup. Time T_1 is recorded, then the mass in the cup (the pulling force) is doubled and time T_2 is recorded. What do you find?**

✔ **ANSWER**

Observation: With double the mass in the cup, the cart reaches the end in less time, so $T_2 < T_1$.

Since s is the same, $a_1/a_2 = T_2^2/T_1^2$, which is less than 1, so $a_2 > a_1$.

Conclusion: For the same mass of the cart, a larger force gives a larger acceleration (acceleration increases as net force increases).

Think as a Scientist

Second hypothesis: for the same force, a smaller mass has a larger acceleration. How can you test it?

✔ ANSWER

Keep the pulling force (mass in the cup) the same and change only the mass of the cart, then compare the times and accelerations. This is done in Activity 6.4.

Activity 6.4

Let us experiment: Activity 6.3 is repeated with the same mass in the cup but double the mass of the cart. Find the ratio of accelerations. What do you find?

✔ ANSWER

Observation: The heavier cart takes more time to cover the same distance, so its acceleration is smaller.

Conclusion: For the same force, increasing the mass decreases the acceleration: acceleration is inversely related to mass.

⭐ Pause and Ponder · 6.5 Newton's Second Law of Motion

6 **A toy car of mass 100 g is moving with a constant velocity of 0.5 m s^{-1}. What is the net force acting on the toy car?**

✔ ANSWER

Constant velocity → acceleration a = 0.

F = ma = 0.1 kg × 0 = **0 N**. No net force acts on the toy car.

7 **Two children of different masses are sitting on identical swings. To impart identical initial acceleration, for which child would you require to apply a larger force? Explain why.**

✔ ANSWER

For the **heavier child**.

F = ma: for the same acceleration a, the force needed is larger when the mass m is larger.

8 **How are glass items packed for transportation using a bubble wrap or hay protected from damage?**

✔ ANSWER

If the box is jerked or dropped, the soft bubble wrap or hay squeezes and lets the glass stop over a **longer time**.

So the acceleration (slowing down) is smaller, and by F = ma the force on the glass is smaller. The glass does not break.

⭐ 6.6 Newton's Third Law of Motion

Activity 6.5 **Let us explore: sitting on a chair with wheels (legs raised), you push a heavy table away and then pull it towards you. What happens to you? What conclusion can you draw?**

✔ ANSWER

Observation: When you push the table forward, your chair moves **backwards** (opposite direction). When you pull the table towards you, your chair moves **towards the table** (forward).

Conclusion: Each time you apply a force on the table, the table applies a force on you in the opposite direction.

Activity 6.6 **Let us verify: two identical spring balances are hooked together; one end is held fixed and the other is pulled. Predict and check the readings. What does it indicate?**

✔ ANSWER

Prediction and observation: Both balances show the **same reading** every time, whatever force you apply.

Conclusion: The forces the two balances apply on each other are equal in magnitude and opposite in direction (Newton's third law).

Activity 6.7 **Let us understand: an inflated balloon is taped to a straw on a taut thread. When its neck is opened, in which direction do the straw and the balloon move?**

✔ ANSWER

Observation: The air rushes out of the neck, and the balloon with the straw moves along the thread in the **opposite direction** (away from the neck).

Conclusion: The balloon pushes the air out; the air pushes the balloon with an equal and opposite force. A rocket moves in the same way.

Pause and Ponder · 6.6 Newton's Third Law of Motion

9 **Why does a fireperson sometimes struggle when holding the pipe issuing water?**

✔ ANSWER

The pipe pushes a large amount of water forward at high speed.

By Newton's third law, the water pushes the pipe **backwards** with an equal force. This strong backward push is hard to hold, so the fireperson has to hold the pipe firmly.

10 **Suppose a spacecraft is moving in a region of space where the gravitational force on it is negligible. Suggest how it can change its velocity.**

✔ ANSWER

By firing its engine (thrusters) to throw out gas in one direction.

The gas pushes the spacecraft in the opposite direction (third law).

- Fire backwards → it speeds up.
- Fire in the direction of motion → it slows down (as Vikram lander of Chandrayaan-3 did).
- Fire sideways → its direction changes.

⭐ Revise, Reflect, Refine

1 **Using a horizontal force F, a table is moved across the floor at a constant velocity. How much is the frictional force exerted by the floor on the table?**

✔ ANSWER

Constant velocity → net force = 0.

So friction balances the applied force: frictional force = **F**, acting opposite to the motion.

2 **For a ball moving on a smooth frictionless surface, choose the correct option:**

(i) If no net force is applied on the ball, the velocity of the ball will remain the same/increase/decrease.

(ii) If a net force is applied on the ball in the direction of its motion, the magnitude of its velocity will remain the same/increase/decrease.

(iii) If a net force is applied opposite to the direction of its motion, the magnitude of its velocity will remain the same/increase/decrease.

✔ ANSWER

(i) **remain the same**

(ii) **increase**

(iii) **decrease**

3 **Two blocks P and Q are on a smooth horizontal surface (Fig. 6.36). Forces of 5 N and 4 N act in opposite directions on block P, while block Q is moving with a constant velocity. Which statement is correct?**

(i) P experiences a net force and Q does not experience a net force.

(ii) P does not experience a net force and Q experiences a net force.

(iii) Both P and Q experience a net force.

(iv) Neither P nor Q experiences a net force.

✔ ANSWER

(i)

Net force on P = 5 N − 4 N = 1 N (towards the 5 N force). Q moves with constant velocity, so net force on Q is zero.

4 **In a snake boat race practice (Vallum kalli in Kerala), 100 oarsmen row a boat. 95 row backwards to propel the boat forward, but by mistake 5 row in the opposite direction. If each oarsman applies a horizontal force of 200 N, what is the net force on the boat? (Ignore drag, air friction, etc.)**

✔ ANSWER

Forward force = 95 × 200 N = 19000 N

Backward force = 5 × 200 N = 1000 N

Net force = 19000 N − 1000 N = **18000 N in the forward direction**

5 **When a net force acts on an object, the object accelerates:**

(i) opposite to the direction of force, with acceleration proportional to the force.

(ii) opposite to the direction of force, with acceleration proportional to the mass.

(iii) in the direction of force, with acceleration inversely proportional to the force.

(iv) in the direction of force, with acceleration proportional to the force.

✔ ANSWER

(iv)

Second law: a = F/m, in the direction of the net force.

6 **Position-time graphs of four objects (Fig. 6.37): A – straight line rising; B – horizontal line; C – curve rising more and more steeply; D – straight line falling. A net force acts on:**

(i) Object A

(ii) Object B

(iii) Object C

(iv) Object D

✔ ANSWER

(iii) Object C

A and D (straight lines) move with constant velocity and B is at rest, so no net force acts on them. C's graph is curved: its velocity keeps increasing, so it accelerates and a net force acts on it.

7 **A sailor jumps out from a small boat to the shore (Fig. 6.38). As the sailor jumps forward, will the boat move? If yes, in which direction and why?**

✔ ANSWER

Yes, the boat moves backwards (away from the shore).

To jump forward, the sailor pushes the boat backwards with his feet. By Newton's third law, the boat pushes him forward with an equal force, and the push on the boat moves it backwards.

8 **During a high jump event, a landing mat or sand bed is placed for the athlete to fall upon (Fig. 6.39). Explain the reason behind it.**

✔ ANSWER

The soft mat or sand lets the athlete stop over a **longer time**.

So the acceleration (slowing down) is smaller and, by F = ma, the force on the body is smaller. This prevents injury.

9 **A hand cart loaded with vegetables collides with an identical but empty hand cart. During the collision:**

(i) the loaded cart exerts a force of larger magnitude on the empty cart.

(ii) the empty cart exerts a force of larger magnitude on the loaded cart.

(iii) neither cart exerts a force on the other.

(iv) the loaded cart and the empty cart, both exert an equal magnitude of force on each other.

✔ ANSWER

(iv)

By Newton's third law, the forces are equal in magnitude and opposite in direction.

10 **The acceleration-mass graph for the acceleration produced by a force on objects of different masses is given (Fig. 6.40). Plot the force-mass graph.**

From the graph: m = 1 kg → a = 10 m s^{-2}; m = 2 kg → a = 5 m s^{-2}; m = 4 kg → a = 2.5 m s^{-2}

✔ ANSWER

F = ma for each point:

Mass (kg)	Acceleration (m s^{-2})	Force F = ma (N)
1	10	10
2	5	10
4	2.5	10

The force is the same, **10 N**, for every mass.

So the force-mass graph is a **straight line parallel to the mass axis** at F = 10 N.

11 **The velocity-time graph of an object of mass 10 kg moving along a straight line is shown (Fig. 6.41): velocity rises steadily from 10 m s^{-1} at t = 0 to 30 m s^{-1} at t = 8 s. Calculate the force acting on the object.**

✔ ANSWER

Given: m = 10 kg, u = 10 m s^{-1}, v = 30 m s^{-1}, t = 8 s

a = (v − u)/t = (30 − 10)/8 = 2.5 m s^{-2}

F = ma = 10 kg × 2.5 m s^{-2}

F = 25 N (in the direction of motion)

12 **A bullet of mass 50 g moving with a speed of 100 m s^{-1} enters a heavy stationary wooden block and stops after penetrating 50 cm. Estimate the stopping force on the bullet (assume constant acceleration).**

✔ ANSWER

Given: m = 50 g = 0.05 kg, u = 100 m s^{-1}, v = 0, s = 50 cm = 0.5 m

$v^2 = u^2 + 2as$

$0 = (100)^2 + 2 \times a \times 0.5$

a = −10000 m s^{-2} (retardation)

F = ma = 0.05 kg × 10000 m s^{-2}

Stopping force = 500 N (opposite to the motion of the bullet)

13 **A footballer kicked a football with a speed of 108 km h^{-1}. The estimated force imparted was 800 N. The mass of the football was 0.4 kg. Calculate the time of contact between the foot and the ball.**

✔ ANSWER

Given: m = 0.4 kg, F = 800 N, u = 0

v = 108 km h^{-1} = 108 × 1000/3600 = 30 m s^{-1}

a = F/m = 800/0.4 = 2000 m s^{-2}

v = u + at → 30 = 0 + 2000 × t

t = 30/2000

t = 0.015 s

14 **An object of mass 2 kg moving with a constant velocity of 10 m s^{-1} enters a rough patch where friction is 7 N. At the same time, an extra constant force of 3 N opposing the motion is applied. How much distance does the object travel before coming to rest?**

✔ ANSWER

Given: m = 2 kg, u = 10 m s^{-1}, v = 0

Total opposing force = 7 N + 3 N = 10 N

a = −F/m = −10/2 = −5 m s^{-2}

$v^2 = u^2 + 2as$ → $0 = (10)^2 + 2 \times (-5) \times s$

s = 100/10

s = 10 m

15 **A tractor pulls a harrow of mass m_1 with a net force F giving acceleration a_1. The same tractor pulls a trolley of mass m_2 with force F giving acceleration a_2. If the tractor now pulls the trolley with the harrow on it (same force F), obtain the resulting acceleration in terms of a_1 and a_2. Ignore friction.**

✔ ANSWER

$m_1 = F/a_1$ and $m_2 = F/a_2$

Total mass = $m_1 + m_2 = F/a_1 + F/a_2 = F(a_1 + a_2)/(a_1a_2)$

$a = F/(m_1 + m_2) = F \times a_1a_2/[F(a_1 + a_2)]$

$a = a_1a_2/(a_1 + a_2)$

16 **When the pole of a bar magnet is brought close to a magnetic compass, the magnet and the compass needle exert equal and opposite magnetic forces on each other. But the compass needle moves, whereas the bar magnet does not (Fig. 6.42). Explain why.**

✔ ANSWER

The forces are equal, but they act on different objects with different masses.

The compass needle is very light and turns freely on its pivot, so the force gives it a noticeable acceleration (a = F/m).

The bar magnet is much heavier and is held in the hand (friction/holding force balances the small force), so it does not move noticeably.

⭐ The Journey Beyond

1 **Does friction depend on how hard the surfaces press each other? Is the friction on an object about to move larger than after motion begins? Is friction on a rolling object less than on a sliding object? Create an infographic.**

💡 SAMPLE ANSWER

- **Yes:** the harder the surfaces press together, the larger the friction.
- **Yes:** friction just before motion starts (static friction) is larger than friction after it starts moving (sliding friction).
- **Yes:** rolling friction is much smaller than sliding friction. That is why the wheel made moving loads much easier.

Show these three points with pictures (box being pushed, heavy and light box, wheel vs sliding block) in your infographic.

2 **Two toy cars of equal mass with bar magnets (like poles touching) are released; distances are recorded, then equal masses are added and the experiment is repeated. Did the cars**

travel equal distances in opposite directions? Plot distance versus mass and discuss.

💡 **SAMPLE ANSWER**

Expected: the two cars move apart in opposite directions and travel (nearly) **equal distances**, because the magnetic forces on them are equal and opposite and their masses are equal. With more mass added, each car travels a **smaller distance** (smaller acceleration for the same force). The distance–mass graph goes down as mass increases.

🎉 **Chapter 6 done! Revise the answers once more tomorrow.**